深圳联讯光电有限公司
SHENZHEN LIANXUN OPTRONICS CO.,LTD

MODEL NO. : LX175A3101CTP01

ISSUED DATE: 2024-12-10

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☐Preliminary Specification
■Final Product Specification

CUSTOMER:

Approved by	Notes

LIANXUN:

PREPARED	CHECKED	APPROVED

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REVISION RECORD

Rev. NO.	Rev. Date	Description	Remarks
1.0	2024-12-10	First Release	

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1. General Specifications

LX175A3101CTP01 is a AMOLED cell. The module display area contains 466x 466pixels. This product accords with RoHS environmental criterion.

Item	Display Panel	Remark
Size	1.75 inch	/
Display Mode	AMOLED	/
Viewing Direction	Free View	/
Module area(W x H x T)	51.5 x 51.5x 1.67	mm
LCM Outline(W x H)	46.35 x 46.35	mm
Active Area(W x H)	43.76(W) x 43.76(H)	mm
Resolution	466RGB x 466	/
Pixel Pitch(W x H)	0.0939 x 0. 0939	mm
Display Drive IC	CO5300	/
Touch Controller IC	CST820	/
Display glass	BOE	/
Interface Type	MIPI&QSPI	/

2. ABSOLUTE MAXIMUM RATINGS

Parameter	Symbol	Min	Max	Unit
Supply voltage (Display)	VCC	-0.3	5.5	V
	IOVCC	-0.3	5.5	V
	ELVDD	3.55	4.65	V
	ELVSS	-2.45	-2.35	V
Operating temperature	TOP	-20	70	°C
Storage temperature	TST	-30	80	°C
Humidity	RH	-	90	%RH

Note: Absolute maximum ratings means the product can withstand short-term, NOT more than 120 hours. If the product is a long time to withstand these conditions, the life time would be shorter.

3. ELECTRICAL CHARACTERISTICS

Parameter		Symbol	Condition	Min	Typ	Max	Unit
Supply voltage (Display)		VCC		3.25	3.3	3.35	V
		IOVCC		1.70	1.8	1.95	V
		ELVDD	-	3.25	3.3	3.35	V
		ELVSS	-	-3.35	-3.3	-3.35	V
Input voltage	'L' level	VIL	IOVCC=1.65V ~3.3V	GND	-	0.2*IOVCC	V
	'H' level	VIH		0.8*IOVCC	-	IOVCC	V
Output voltage	'L' level	VOL	I(OH)=-1mA I(OL)=+1mA	GND	-	0.2*IOVCC	V
	'H' level	VOH		0.8*IOVCC	-	IOVCC	V
Current (Display)	Sleep out mode	IVCC	Full white display (450nits)	-	6	7.2	mA
		IIOVCC		-	3	3.6	mA
		IELVDD/ELVSS		-	22	25.3	mA
	Sleep in mode	IVCC		-	20	40	uA
		IIOVCC		-	50	100	uA
	Deep Standby Mode	IVCC	Power Consumption	-	-	<50	uW
		IIOVCC		-	-	<50	uW
Frame Frequency		fFRM		-	60	-	Hz

4. RECOMMENDED OPERATION SEQUENCE

Power on sequence

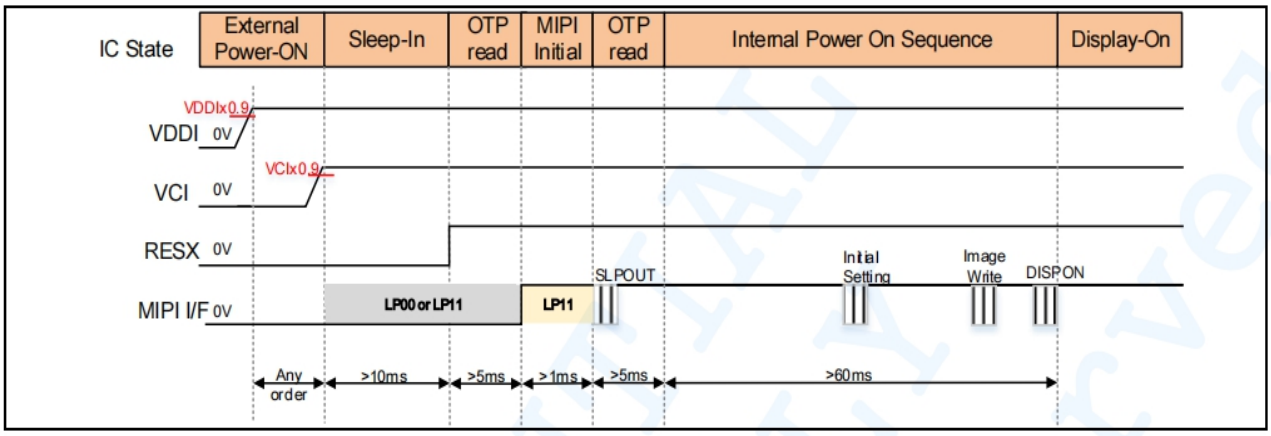


Figure 15 Power-On Sequence

Power off sequence

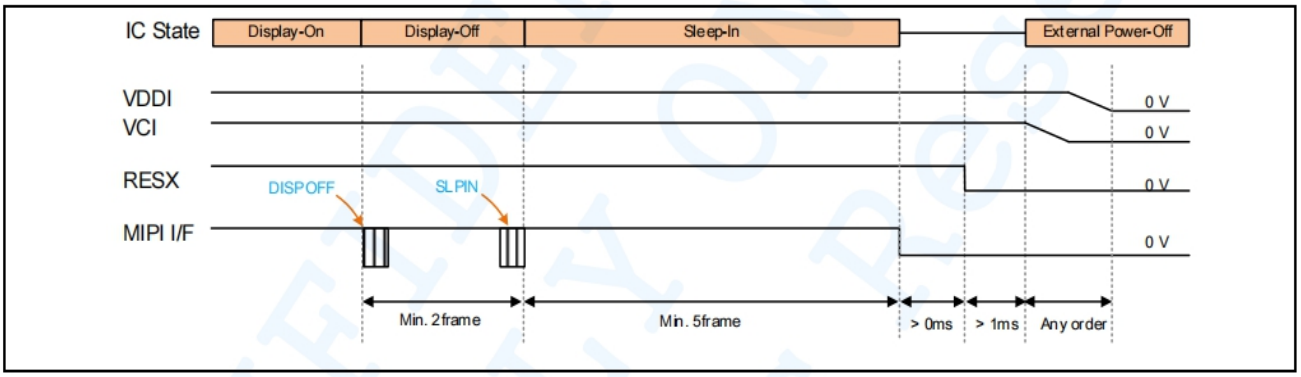


Figure 16 Power-Off Sequence

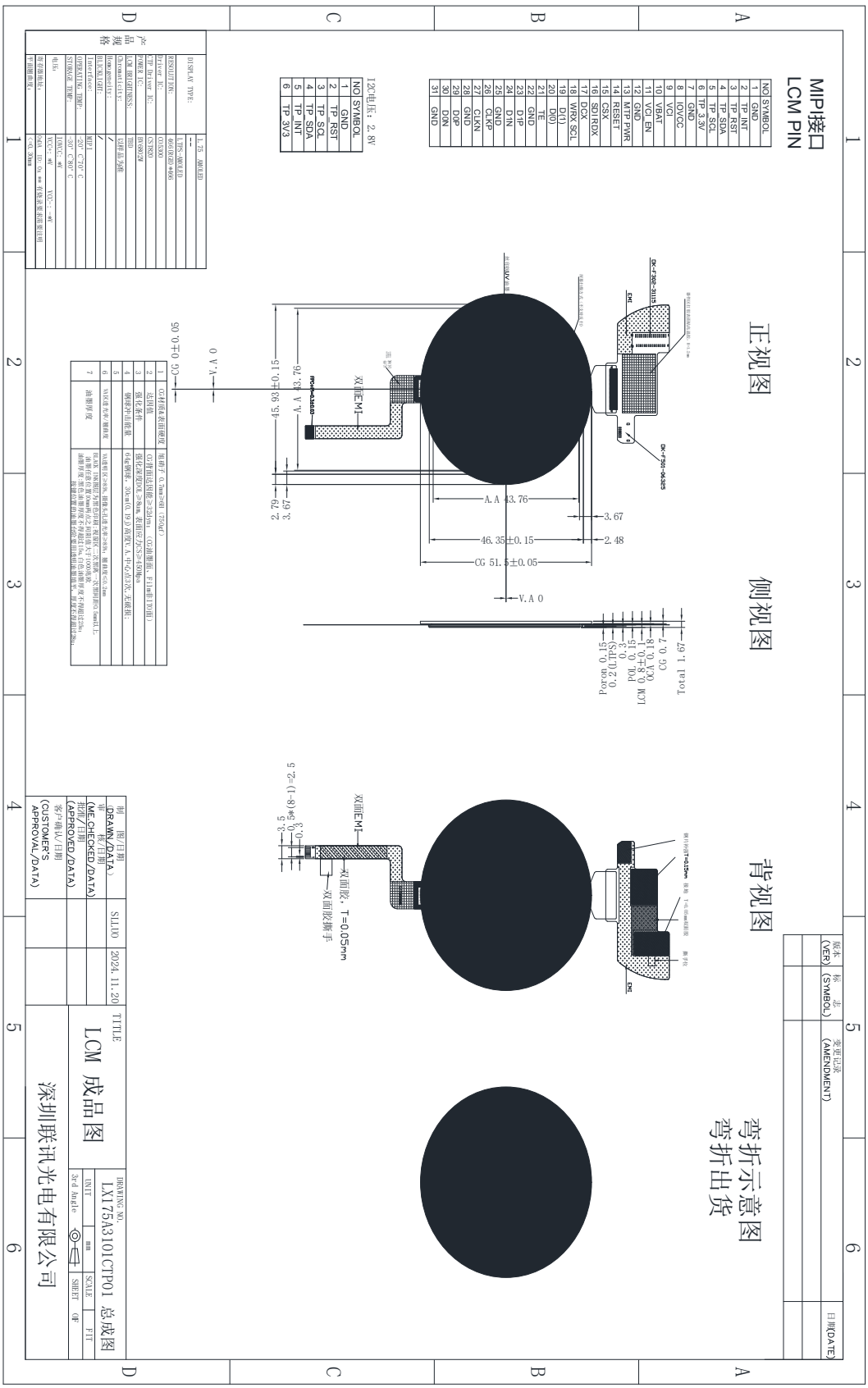
Reset Timing

Reset input timing:

IOVCC=1.65 to 3.3V，VCC=2.7 to 3.6V，AGND=DGND=0V，Ta=-40 to 85℃

Symbol	Parameter	Related Pins	MIN	TYP	MAX	Note	Unit
tRESW	Reset low pulse width	RESX	10	-	-	-	Us
tREST	Reset complete time	-	-	-	10	When reset applied during Sleep in mode	ms
		-		-	120	When reset applied during Sleep out mode	ms

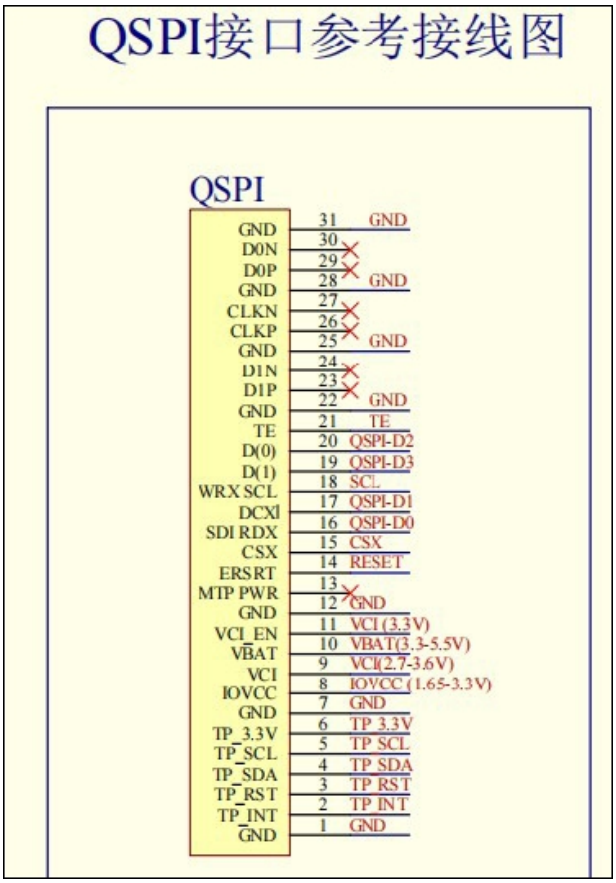
5. DIMENSIONAL DRAWING

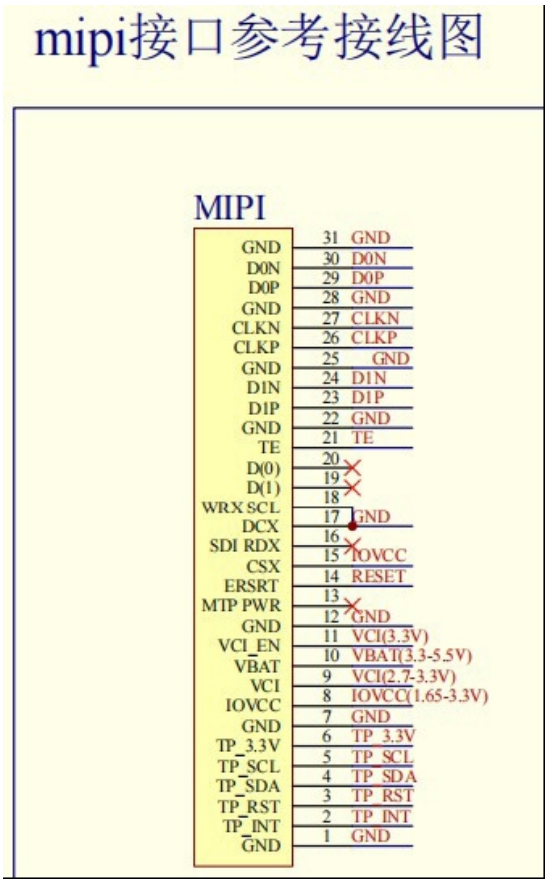


6. INTERFACE PIN CONNECTIONS

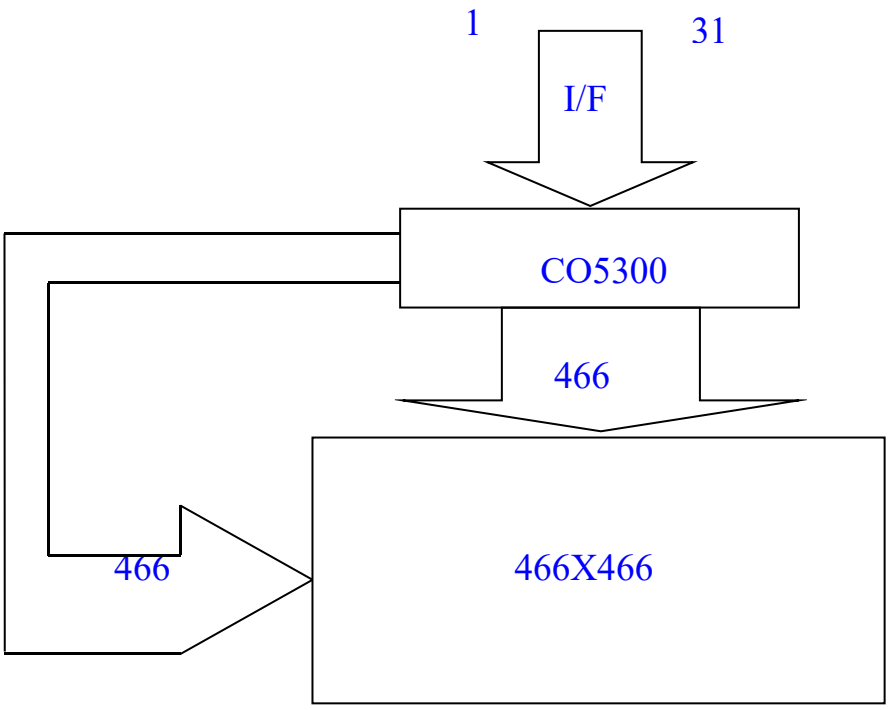
Pin.No	Symbol	Function
1	GND	Ground.
2	TP_INT	TP PIN
3	TP_RST	TP PIN
4	TP_SDA	TP PIN
5	TP_SCL	TP PIN
6	TP_3V3	TP PIN(2.8-3.3V)
7	GND	Ground.
8	IOVCC	Power supply (1.8-3.3V)
9	VCI	Power supply(3.3V)
10	VBAT	Power supply.(3.7-4.5V)
11	VCI_EN	Enable for VBAT
12	GND	Ground
13	MTP_PWR	Power supply for OTP.
14	RESET	This signal will reset the device and it must be applied to properly initialize the chip.
15	CSX	Chip select signal input(low active).
16	SDI RDX	serial data input 0 / serial data output
17	DCX	serial data input 1
18	WRX SCL	This pin is used serial interface clock in 3-wire 9-bit / 4-wire 8-bit /QSPI serial interface.
19	D(1)	serial data input 3
20	D(0)	serial data input 2
21	TE	Tearing effect output pin to synchronize MPU to frame writing.
22	GND	Ground
23	D1P	MIPI-DSI Data differential signal input pins.
24	D1N	MIPI-DSI Data differential signal input pins.
25	GND	Ground
26	CLKP	CLKP
27	CLKN	CLKN
28	GND	Ground
29	D0P	MIPI-DSI Data differential signal input pins.
30	D0N	MIPI-DSI Data differential signal input pins.
31	GND	Ground

Application circuit





7. BLOCK DIAGRAM OF LCM



8. ELECTRO-OPTICAL CHARACTERISTICS

Optical characteristics are determined after the unit has been on and stable for approximately 30 minutes dark environment at 25℃.the value specified are at an approximate distance 500mm from the LCD surface at a viewing angle and θ equal to 0

8.1 LCD Optical Characteristics

Item	Symbol	Condition	Min	Type	Max	Unit	Note
Surface Luminance	Lv	θ=0° ∅=0° Ta=25℃	-	700	-	cd/m2	Note1
Luminance uniformity	δ WHITE		80	85	-	%	Note2
Contrast Ratio	Cr		100000	-	-	-	Note3
Viewing Angle	θ	Up/Down/Right/Left Cr≥10	80	-	-	deg	Note4
Color Coordinate of CIE1931	Red x	θ=0° ∅=0° Ta=25℃.	0.636	0.666	0.696	-	Note 5
	Red y		0.303	0.333	0.363	-	
	Green x		0.186	0.226	0.266	-	
	Green y		0.679	0.719	0.759	-	
	Blue x		0.118	0.138	0.158	-	
	Blue y		0.035	0.055	0.075	-	
	White x		0.280	0.300	0.320	-	
	White y		0.290	0.310	0.330	-	
NTSC ratio	NTSC	CIE1931	-	100	-	%	-
Cross talk	-	25℃.	-	-	1.1	%	-
Flicker	-	25℃.		-	-30	DB	-
Gamma	-	–	1.9	2.2	2.5	-	
Lifetime(450nits)	T95	25℃	150	240		h	

Note 1. Surface Luminance

Measurement equipment: CS2000 or similar equipment.

Measuring surroundings: Dark room.

Measuring temperature: Ta=25℃.

The data are measured after OLEDs are lighted on for more than 5 minutes and all pixels are fully white.

The Surface Luminance is the average value of 5 measured spots (Fig-1):

Lv = Average Luminance with all white pixels (P_1, P_2, P_3, P_4, P_5)

Note 2. Luminance Uniformity

Measurement equipment: CS2000 or similar equipment.

Measuring surroundings: Dark room.

Measuring temperature: Ta=25℃.

The data are measured after OLEDs are lighted on for more than 5 minutes and all pixels are fully white.

The Luminance Uniformity is calculated by using following formula:

WHITE = Lp (Min.) / Lp (Max.)×100 (%)

Lp (Min.) = Minimum Luminance with all white pixels (P1, P2, P3, P4, P5) Lp (Max.) = Maximum Luminance with all white pixels (P1, P2, P3, P4, P5)

Note 3. Contrast Ratio(CR)

Measurement equipment: CS2000 or similar equipment.

Measuring surroundings: Dark room.

Measuring temperature: Ta=25℃.

The data are measured after OLEDs are lighted on for more than 5 minutes.

The Contrast Ratio is calculated by using following formula:

Contrast Ratio(Cr) = Lw / Lb

Lw = Average Luminance with all **white** pixels (P_1, P_2, P_3, P_4, P_5) Lb = Average Luminance with all **black** pixels (P_1, P_2, P_3, P_4, P_5)

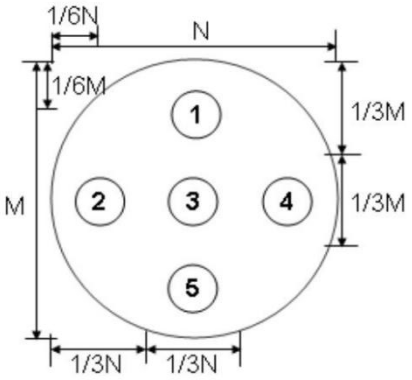


Fig-1

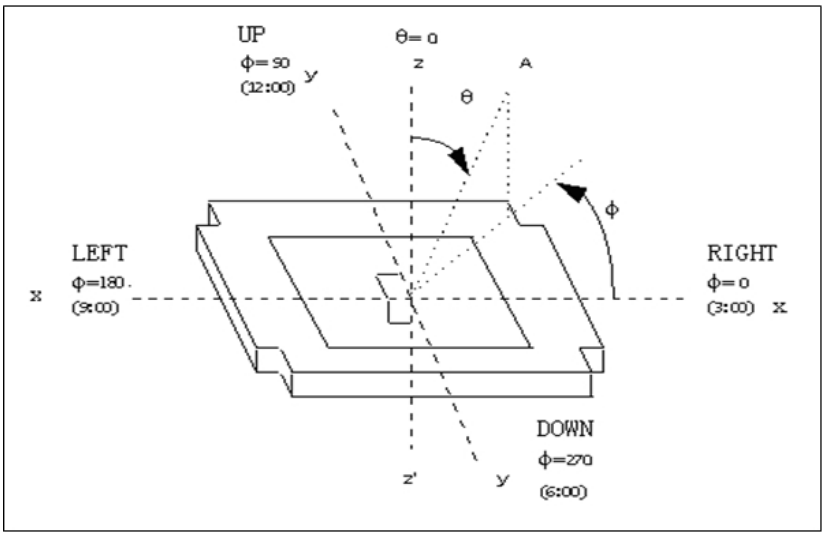
Note 4. Viewing Angle

Measurement equipment: DMS803 or similar equipment.

Measuring surroundings: Dark room.

Measuring temperature: Ta=25℃.

The Viewing Angle is the angle at which the contrast ratio is greater than 10. The angles are determined for the horizontal or x axis and the vertical or y axis with respect to the z axis which is normal to the display surface.



Note 5. Color Coordinate of CIE1931

Measurement equipment: CS2000 or similar equipment.

Measuring surroundings: Dark room.

Measuring temperature: Ta=25℃.

The x, y value of Color Coordinate is determined by measuring at center position of the display panel.

9. TOUCH PANEL SPECIFICATION

9.1 General Specification

Item	Display Panel	Remark
Lens	旭硝子	/
Outline Dimension	51.5 x 51.5 x 0.7	mm
View Area	44.16 x 44.16	mm
Touch Panel Structure	ONCELL	/
Surface Hardness	≧7H	/
Controller IC	CST820	/
Interface Type	IIC	/

9.2 FPC PIN Assignment

Pin.No	Symbol	Function
1	GND	Power Ground
2	TP_RST	Reset pin for the TP
3	TP_SCL	IIC clock for the data input
4	TP_SDA	IIC Data input
5	TP_INT	Interrupt for the TP
6	TP_3V3	Power Supply

10.0 INSPECTION CRITERIA

10.1 Inspection Conditions

10.1.1 Environmental conditions

The environmental conditions for inspection shall be as follows

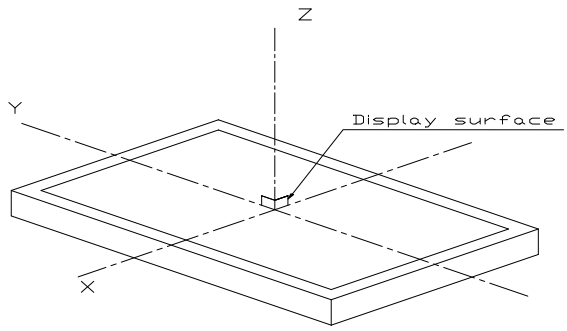
Room temperature: 20±3°C

Humidity: 65±20%RH

10.1.2 Environmental conditions

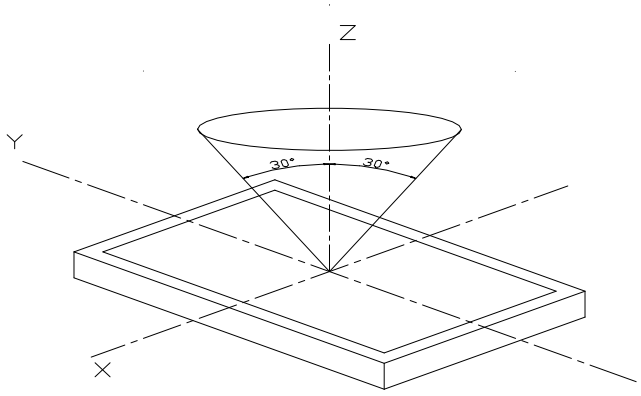
With a single 20-watt fluorescent lamp as the light source, the inspection was in the distance of 30cm or more from the LCD to the inspector's eyes.

10.2 Light Method



Fluorescent lamp perpendicular to the display surface

Inspection distance and angle



Inspection should be performed within angle Φ (Φ is usually 30°) from Z axis to each X and Y.

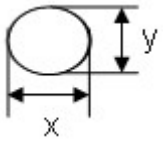
Inspection distance in any direction within Φ must be kept 30±5cm from the display surface.

10.3 Classification of defects

10.3.1 Major defect

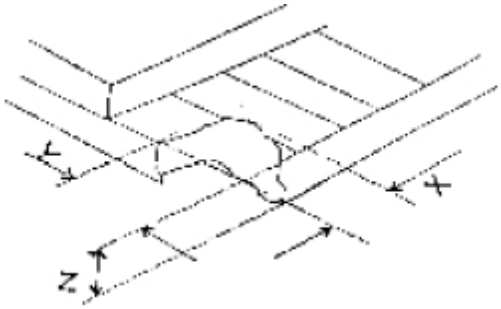
No.	Item	Inspection Standard	Classification of defects
1	All functional defects	1) No display 2) Display abnormally 3) Open or missing segment 4) Short circuit 5) Excess power consumption 6) Backlight no lighting, flickering and abnormal lighting	Major
2	Missing	Missing component	Major
3	Outline dimension	Overall outline dimension beyond the drawing is not allowed	Major

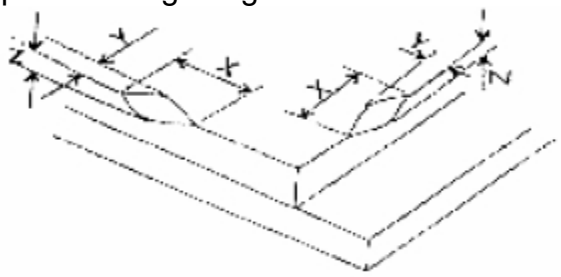
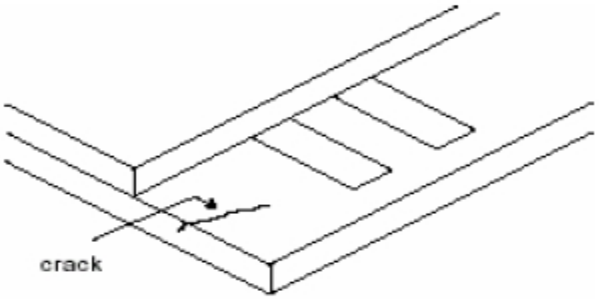
10.3.2 Cosmetic Defect

<table>
<tr><th>No.</th><th>Item</th><th colspan="2">Inspection Standard</th><th>Classification of defects</th></tr>
<tr><td rowspan="2">1</td><td rowspan="2">(spot defect)
Black and White spot pinhole</td><td>For dark/white spot, size Φ is defined as Φ=(x+y)/2</td><td> 
</td><td rowspan="2">Minor</td></tr>
<tr><td colspan="2"><table>
<tr><th>Size Φ (mm)</th><th>Acceptable Quantity</th></tr>
<tr><td>Φ≤0.1</td><td>Ignore</td></tr>
<tr><td>0.10<Φ≤0.15</td><td>2</td></tr>
<tr><td>0.15<Φ≤0.2</td><td>1</td></tr>
<tr><td>0.2<Φ</td><td>0</td></tr>
</table></td></tr>
<tr><td>2</td><td>(line defect)
Black and White line</td><td colspan="2">Define: 
</td><td>Minor</td></tr>
</table>

<table>
<tr><td></td><td>Polarizer scratch</td><td><table><tr><th>Width(mm)</th><th>Length(mm);Acceptable Qty</th></tr><tr><td>Φ≤0.03 L≤1.0</td><td>Ignore</td></tr><tr><td>0.03<W≤0.05</td><td>L≤3.0; N≤2</td></tr><tr><td>0.05<W or L>3.0</td><td>Define as spot defect</td></tr></table></td><td></td></tr>
<tr><td>3</td><td>Polarizer defect</td><td>Dent or bubble(between the polarizer and glass)<table><tr><th>Size Φ(mm)</th><th>Acceptable Qty</th></tr><tr><td>Φ≤0.10</td><td>Ignor</td></tr><tr><td>0.10<Φ≤0.15</td><td>2</td></tr><tr><td>0.15<Φ≤0.20</td><td>1</td></tr><tr><td>0.20<Φ</td><td>0</td></tr></table></td><td>Minor</td></tr>
</table>

10.3.3 Cosmetic Defect

<table>
<tr><th>No.</th><th>Item</th><th>Inspection Standard</th><th>Classification of defects</th></tr>
<tr><td>1</td><td>Glass defect</td><td>1) Chip on the corner
 
<table><tr><th>X</th><th>Y</th><th>Z</th></tr><tr><td>≤3.0</td><td>≤S</td><td>≤T</td></tr></table>Remark: S=contact pad length;
T=the thickness of glass
Chips on the corner of terminal shall not be allowed to extend into the ITO pad or expose perimeter seal. Acceptable Quantity N≤2.</td><td>Minor</td></tr>
</table>

<table>
<tr>
<td rowspan="2"></td>
<td rowspan="2"></td>
<td>2) Chip on the edge of glass
 
<table>
<tr><th>X</th><th>Y</th><th>Z</th></tr>
<tr><td>Ignore</td><td>≤0.5</td><td>≤T</td></tr>
</table>
Acceptable Quantity: N≤2</td>
<td>Minor</td>
</tr>
<tr>
<td>3) Creak
Creaks tend to break are not allowed.
 
</td>
<td>Minor</td>
</tr>
</table>

11 . ELIABILITY AND INSPECTION STANDARD

NO.	Test Item	Test condition	Criterion
1	High Temperature Storage	80℃±3℃ 24H Restore 2H at 25℃ Power off	After testing, cosmetic and electrical defects should not happen.
2	Low Temperature Storage	-30℃±3℃ 24H Restore 2H at 25℃ Power off	
3	High Temperature Operation	70℃±3℃ 24H Restore 2H at 25℃ Power on	
4	Low Temperature Operation	-20℃±3℃ 24H Restore 2H at 25℃ Power on	
5	High Temperature & Humidity Storage	50℃±3℃ 80±3%RH 24H Power off	
6	Temperature Cycle	-20℃←→25℃←→70℃ 30min 5min 30min after 10cycle, Restore 2H at 25℃ Power off	
7	Vibration Test	Frequancy10Hz~55Hz~10Hz Ampliute:1.5mm,X,Y,Zdirection total 1H;	
8	ESD Test	±8KV,Air Mode,150pF/330Ω；	
9	Drop Test(package state)	600mm, concrete floor,1corner, 3edges, 6 sides each time	1.After testing, cosmetic and electrical defects should not happen. 2.The product should remain at initial place. 3.Product uncovered or package broken is not permitted.

12. PRECAUTIONS FOR USING LCD MODULE

12.1 Handing Precautions

(1) The display panel is made of glass and polarizer. As glass is fragile, it tends to become or chipped during handling especially on the edges. Please avoid dropping or jarring. Do not subject it to a mechanical shock by dropping it or impact.

(2) If the display panel is damaged and the liquid crystal substance leaks out, be sure not to get any in your mouth. If the substance contacts your skin or clothes, wash it off using soap and water.

(3) Do not apply excessive force to the display surface or the adjoining areas since this may cause the color tone to vary. Do not touch the display with bare hands. This will stain the display area and degraded insulation between terminals (some cosmetics are determined to the polarizer).

(4) The polarizer covering the display surface of the LCD module is soft and easily scratched. Handle this polarizer carefully. Do not touch, push or rub the exposed polarizers with anything harder than an HB pencil lead (glass, tweezers, etc.). Do not put or attach anything on the display area to avoid leaving marks on. Condensation on the surface and contact with terminals due to cold will damage, stain or dirty the polarizer. After products are tested at low temperature they must be warmed up in a container before coming is contacting with room temperature air.

(5) If the display surface becomes contaminated, breathe on the surface and gently wipe it with a soft dry cloth. If it is heavily contaminated, moisten cloth with one of the following solvents
- Isopropyl alcohol
- Ethyl alcohol
Do not scrub hard to avoid damaging the display surface.

(6) Solvents other than those above-mentioned may damage the polarizer. Especially, do not use the following.
- Water
- Ketone
- Aromatic solvents
Wipe off saliva or water drops immediately, contact with water over a long period of time may cause deformation or color fading. Avoid contacting oil and fats.

(7) Exercise care to minimize corrosion of the electrode. Corrosion of the electrodes is accelerated by water droplets, moisture condensation or a current flow in a high-humidity environment.

(8) Install the LCD Module by using the mounting holes. When mounting the LCD module make sure it is free of twisting, warping and distortion. In particular, do not forcibly pull or bend the I/O cable or the backlight cable.

(9) Do not attempt to disassemble or process the LCD module.

(10) NC terminal should be open. Do not connect anything.
(11) If the logic circuit power is off, do not apply the input signals.
(12) Since LCM has been assembled and adjusted with a high degree of precision, avoid applying excessive shocks to the module or making any alterations or modifications to it.
- Do not alter, modify or change the shape of the tab on the metal frame.
- Do not make extra holes on the printed circuit board, modify its shape or change the positions of components to be attached.
- Do not damage or modify the pattern writing on the printed circuit board.
- Absolutely do not modify the zebra rubber strip (conductive rubber) or heat seal connector.
- Except for soldering the interface, do not make any alterations or modifications with a soldering iron.
- Do not drop, bend or twist LCM.

12.2 Storage Precautions

When storing the LCD modules, the following precaution is necessary.
(1) Store them in a sealed polyethylene bag. If properly sealed, there is no need for the dessicant.
(2) Store them in a dark place. Do not expose to sunlight or fluorescent light, keep the temperature between 0°C and 35°C.
(3) The polarizer surface should not come in contact with any other objects. (We advise you to store them in the container in which they were shipped).

12.3 Others

Liquid crystals solidify under low temperature (below the storage temperature range) leading to defective orientation or the generation of air bubbles (black or white). Air bubbles may also be generated if the module is subject to a low temperature.
If the LCD modules have been operating for a long time showing the same display patterns, the display patterns may remain on the screen as ghost images and a slight contrast irregularity may also appear. A normal operating status can be regained by suspending use for some time. It should be noted that this phenomenon does not adversely affect performance reliability.